EDWARD VARVAK

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EDUCATION

University of Chicago *Sept 2026 - May 2027*
Doctor of Philosophy (Ph.D.) in Mathematics, advised by Ben Bakker

University of Illinois Chicago *Sept 2022 - Sept 2026 (transfer)*
Doctor of Philosophy (Ph.D.) in Mathematics, advised by Ben Bakker,

Stony Brook University *Sept 2018 - May 2022*
Bachelor of Science (B.S.) in Mathematics
Bachelor of Science (B.S.) in Applied Mathematics and Statistics

RESEARCH INTERESTS

Complex algebraic geometry, Hodge theory, Abelian varieties, symplectic varieties, Lagrangian fibrations, algebraic foliations, monodromy representations, non-Abelian Hodge theory.

RESEARCH AND PUBLICATIONS

3. *The Mumford–Tate group of a very general Lagrangian fiber*, 2026.
2. *The Monodromy of Compact Lagrangian Fibrations*, 16 pages, **IMRN**, rnag057, 2026.
1. *Recovering Plane Nodal Curves from their Bitangents*, 12 pages, **Undergraduate Honors Thesis**, advised by Olivier Martin, 2022.

INVITED TALKS

4. *Special Session on Higgs Bundles, Character Varieties, and Moduli Spaces*, 2027 Joint Math Meeting in Chicago — *January 2027*
3. *GTA Philadelphia*, Temple University — *May 2026*
2. *Research Seminar*, Institute for Advanced Study — *November 2025*
1. *Graduate Student Colloquium*, University of Illinois Chicago — *February 2025*

ORGANIZATION AND SERVICE

4. Co-organized Graduate Algebraic Geometry Seminar (GAGS) — *Fall 2025-Spring 2026*
3. Judged for QED Compitition (Math Circles of Chicago) — *December 2025*
2. Organized Weil Conjecture Reading Group — *Fall 2025*
1. Organized Lie Groups Seminar — *Fall 2024*

HONORS AND AWARDS

Bernard Kurtin Award for Research in Mathematics, UIC MSCS Department — *2025-2026*
UIC RTG Grant — *Fall 2025, Summer 2025, Spring-Summer 2024, Spring-Summer 2023*

Honors in Mathematics, SBU Mathematics Department *2018-2022*
Kuga-Sah Award for Excellence in Mathematics, SBU Mathematics Department *2022*
Cum Laude, Stony Brook University *2022*

TEACHING EXPERIENCE

University of Chicago:
Math 15200: Calculus II (Graduate Student Lecturer) *Fall 2026*

University of Illinois Chicago:
Math 181: Calculus II (Teaching Assistant) *Spring 2026*
Math 160: Finite Math for Business (Teaching Assistant) *Spring 2025, Fall 2024, Fall 2023, Fall 2022*

LANGUAGES

English (Native)
Russian (Fluent)

REFERENCES

Benjamin Bakker (Doctoral Advisor)
Professor
University of Chicago
bakker.uchicago@gmail.com

Valentino Tosatti
Professor
Courant Institute
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İzzet Coşkun
LAS Distinguished Professor
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McKinley Meyer (teaching)
Lecturer
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